Practical No. 9

Aim: Programming threads, handles and asynchronized programs.

build.gradle.kts (Module: app)

```
dependencies {
    ...
    implementation ("org.jetbrains.kotlinx:kotlinx-coroutines-android:1.7.3")
}
```

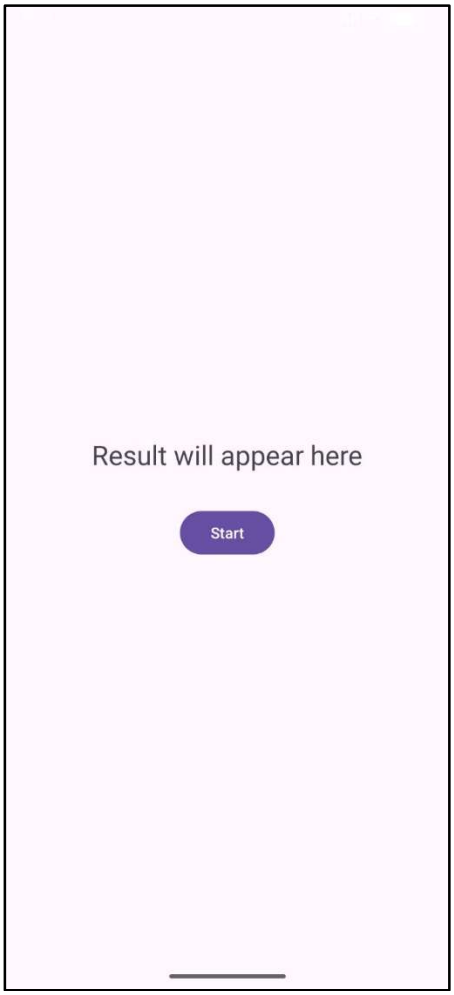
activity_main.xml

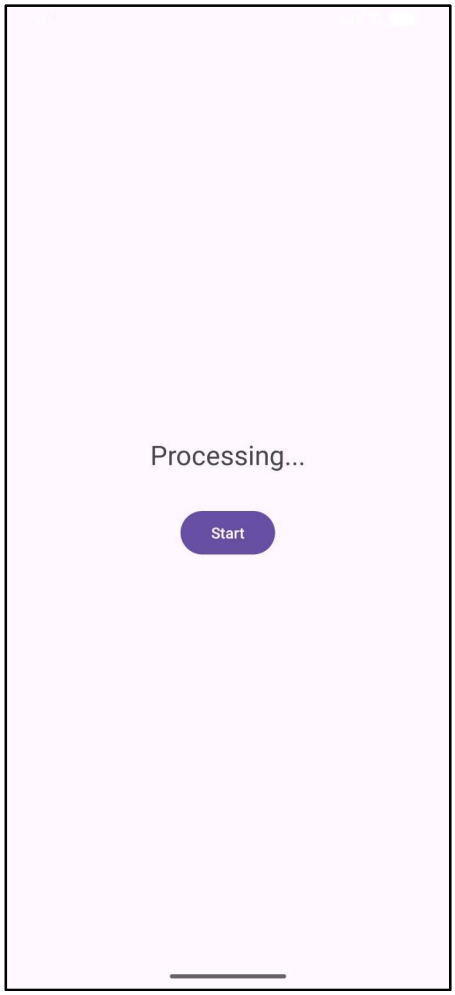
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

     <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="25sp" />

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="Start" />

</LinearLayout>
```

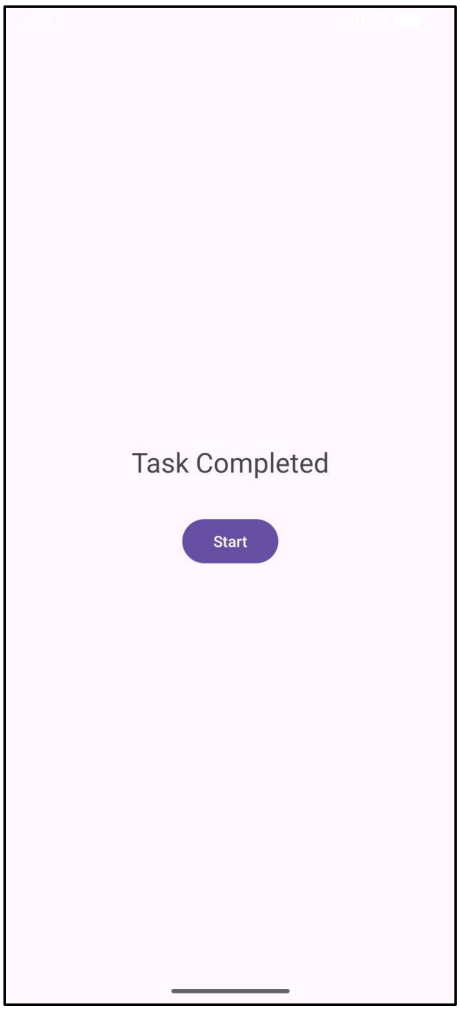
MainActivity.kt

```
package com.example.test

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import kotlinx.coroutines.*

class MainActivity : AppCompatActivity() {

    private lateinit var resultTextView: TextView
    private lateinit var startButton: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        resultTextView = findViewById(R.id.tvResult)
        startButton = findViewById(R.id.btnStart)

        startButton.setOnClickListener {
            // Launch a coroutine on the Main (UI) thread
            CoroutineScope(Dispatchers.Main).launch {
                resultTextView.text = "Processing..."
                val result = withContext(Dispatchers.IO) {
                    // Background task simulation
                    delay(3000)
                    "Task Completed"
                }
                resultTextView.text = result // Update UI
            }
        }
    }
}
```

Output


Result will appear here
Start


Processing...
Start


Task Completed
Start